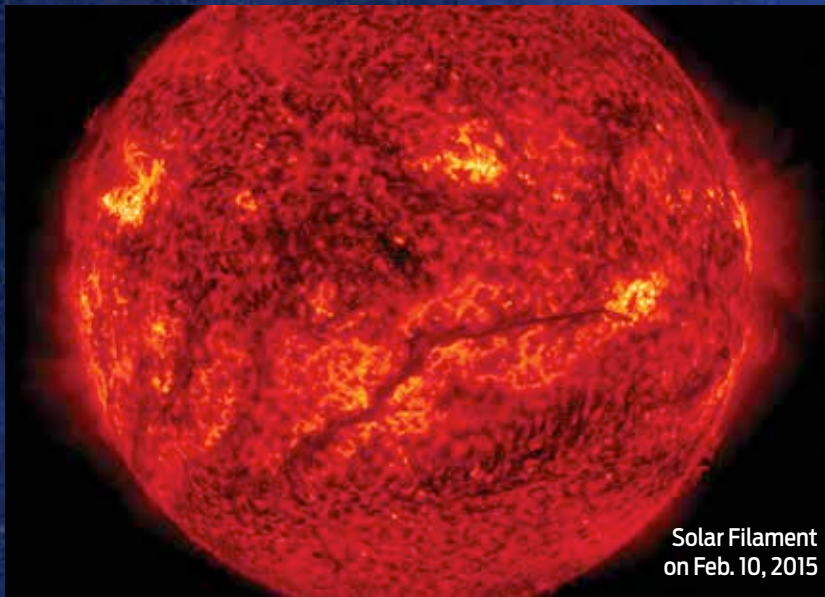


Source: NASA



Solar Filament
on Feb. 10, 2015

The Solar Storm Regiment

Detecting and studying the cosmic frameworks and the impact on our planet

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Right now, solar activity is at a record-breaking level. It is due to the fact that Sun's approaching solar cycle's 11-year high, which occurs in the latter part of this interim phase before things return to normal. This is the primary cause of the Sun's frequent high density of sunspots and magnetic filaments, all of which are signs of unstable magnetic fields. While the National Oceanic and Atmospheric Administration (NOAA) attempts to develop a model on the CME (Coronal Mass Ejections) strength, NASA SOHO has provided the earliest forecast.

The resulting geomagnetic storm will reach G2-class levels, according to SOHO. Such potent solar storms

frequently produce stunning aurora displays. But don't let the flashing lights deceive you; they can seriously damage our digital infrastructure. These solar storms have the ability to corrupt sensitive ground-based electronics, disrupt cellular and internet connections, crash power grids, and harm satellites.

What are Solar Storms?

A solar flare is a powerful radiation flash caused by an explosion of magnetic energy from a sunspot. The greatest explosive occurrences in our solar system are flares. They can persist for a few minutes to several hours and can be observed as bright spots in the Sun. A solar flare is often visible

through the tiny particles or photons it emits at nearly all wavelengths. X-rays and optical light are the main methods used to track flares. Electrons, protons, and massive particles are also accelerated at flares.

Strong magnetic fields shape the corona, the outer solar atmosphere. The limited solar atmosphere can abruptly and powerfully expel pockets of gas and magnetic fields known as coronal mass ejections, where these forces remain closed, frequently over sunspot groupings. One billion tonnes of matter, which can be propelled to several million miles per hour in a dramatic explosion, can be found in a massive CME. Through the intergalactic medium, solar material flows out, striking any planets or spacecraft in the vicinity. CMEs can happen independently but are occasionally linked to flares. Massive accumulations of plasma and magnetic fields that explode from the Sun are known as coronal mass ejections or CMEs. These clouds have the ability to explode in any direction, then move forward in that direction while cutting across the solar wind. The CME won't hit Earth, and so produce impacts unless the cloud is pointed towards it.

Why study them?

Solar System's most potent explosions are solar flares. They directly affect Earth's atmosphere. Here's how:

- The ionization and expansion of the planet's upper atmosphere increases.
- The ensuing alteration in the Earth's ionosphere can cause long-distance radio communications to be interfered with.
- The increased drag caused by the expanded atmosphere can cause the trajectory of satellites around the Earth to be altered.
- Electronic parts of satellites are susceptible to damage.

Astronauts and technological equipment in space are at risk from